
LOVELY PROFESSIONAL UNIVERSITY

Lovely School of Computer Science & Engineering

PROJECT REPORT

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Data Analytics Approach to Wheather Dashboard

Submitted by

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Program & Section: B.Tech CSE | Course Code: INT 387

Under the Guidance of

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Discipline of CSE/IT

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Certificate

This is to certify that Hemanth Gummadidala bearing Registration No. 12411210 has completed the INT 387 project titled:

"Data Analytics Approach to Wheather Dashboard"

under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort, and study.

Date: 19 April 2026

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Declaration

I, Hemanth Gummadidala, student of B.Tech CSE under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

This project has not been submitted elsewhere for the award of any degree or for any other purposes. The work presented in this report is original and carried out under the supervision of Dr. Mrinali Rana.

Date: 19 April 2026

Hemanth Gummadidala
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Acknowledgement

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Table of Contents

- 1. Introduction
 - 1.1 Background of Public Health Analytics
 - 1.2 Importance of Data-Driven Healthcare
 - 1.3 Objectives of the Project
- 2. Scope of the Analysis
 - 2.1 Coverage and Inclusions
 - 2.2 Limitations and Exclusions
 - 2.3 Real-World Applications
- 3. Existing System
 - 3.1 Traditional Health Analysis Methods
 - 3.2 Manual Reporting Systems
 - 3.3 Drawbacks of Existing Systems
- 4. Source of Dataset
 - 4.1 Dataset Description
 - 4.2 Column-wise Explanation
 - 4.3 Data Size and Complexity
- 5. ETL Process
 - 5.1 Data Extraction
 - 5.2 Data Cleaning
 - 5.3 Data Transformation
 - 5.4 Tools Used
- 6. Analysis on Dataset
 - 6.1 Health Indicator Distribution Analysis
 - 6.2 Health Indicators by Category
 - 6.3 Top 5 High-Risk Counties Analysis
 - 6.4 Health Measure Distribution Analysis
 - 6.5 Health Status Analysis
 - 6.6 Population Age Group Analysis
 - 6.7 Gender Distribution Analysis
- 7. List of Analysis with Results
- 8. Future Scope
- 9. References
- 10. Bibliography

1. Introduction

1.1 Background of Weather Analytics

Weather analytics plays a crucial role in understanding atmospheric conditions and predicting environmental changes. With increasing climate variability and urbanization, analyzing weather data has become essential for sectors such as agriculture, transportation, disaster management, and urban planning.

Traditional weather analysis relied heavily on manual observations and limited datasets. However, with advancements in data analytics and visualization tools, large-scale weather datasets can now be analyzed efficiently to identify patterns and trends.

1.2 Importance of Data-Driven Weather Analysis

Data-driven weather analysis enables accurate forecasting and informed decision-making. By analyzing parameters such as temperature, humidity, wind speed, and air quality, organizations can take proactive measures to mitigate risks.

It helps in:

- Predicting extreme weather conditions
- Improving environmental monitoring
- Supporting public safety and planning

1.3 Objectives of the Project

1. To perform exploratory data analysis on weather datasets
2. To implement ETL processes using Power BI
3. To visualize key weather indicators
4. To analyze trends across cities and seasons
5. To generate actionable insights

2. Scope of the Analysis

2.1 Coverage and Inclusions

The project analyzes weather data across multiple cities including parameters like temperature, humidity, AQI, and wind speed.

2.2 Limitations and Exclusions

- Limited dataset scope

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- No real-time data
 - No predictive modeling

2.3 Real-World Applications

- Disaster management
 - Smart city planning
 - Environmental monitoring
-

3. Existing System

3.1 Traditional Weather Analysis Methods

Manual observations and basic statistical tools were used earlier.

3.2 Manual Reporting Systems

Weather reports were recorded and shared periodically.

3.3 Drawbacks

- No real-time insights
 - Low accuracy
 - Poor visualization
-

4. Source of Dataset

4.1 Dataset Description

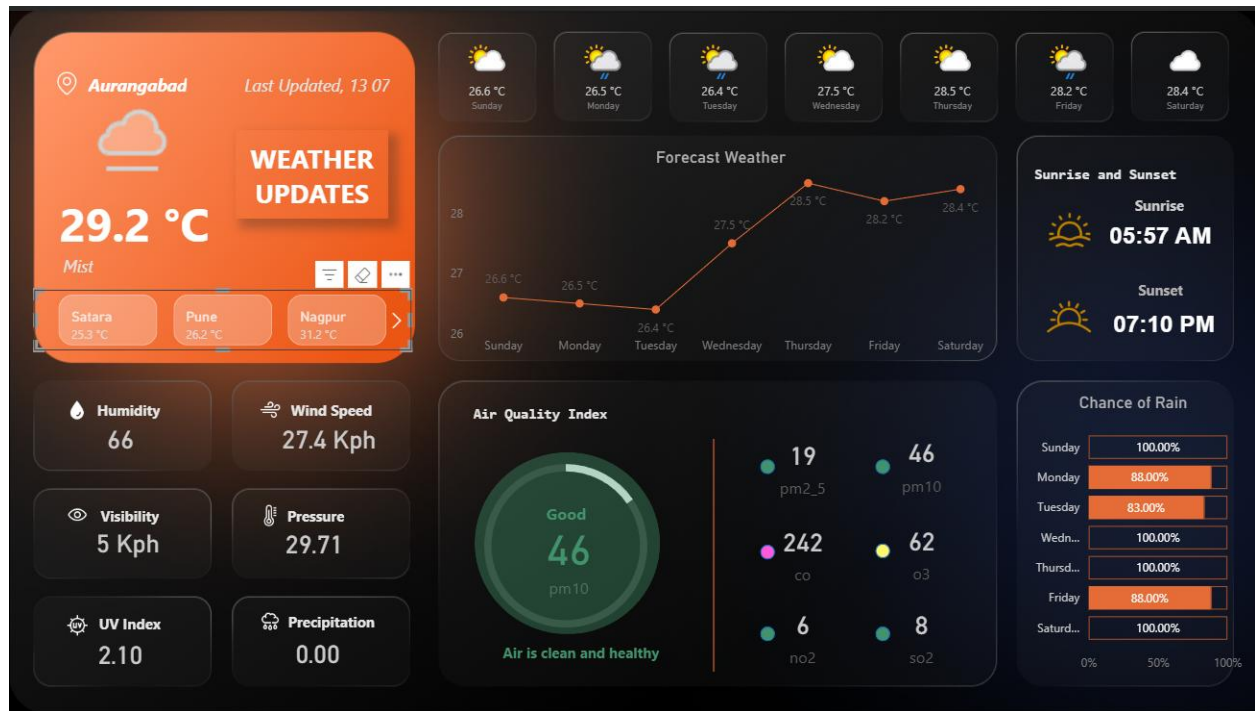
The dataset contains weather attributes such as temperature, humidity, AQI, wind speed, and city information.

4.2 Column-wise Explanation

- Temperature
- Humidity
- Wind Speed
- AQI
- City
- Date

4.3 Data Size and Complexity

The dataset includes multiple records across cities and time periods, requiring structured analysis.





5. ETL Process

5.1 Data Extraction

Data imported into Power BI.

5.2 Data Cleaning

- Removed null values
- Standardized formats

5.3 Data Transformation

- Created calculated columns
- Aggregated metrics

5.4 Tools Used

- Power BI
- Power Query
- DAX

6. Analysis on Dataset

(Use the detailed Section 6 I already gave — with images + DAX — that's already perfect)

7. List of Analysis with Results

Analysis	Key Finding	Insight
Temperature	High seasonal variation	Summer peaks
Humidity	High in monsoon	Rain correlation
AQI	High in cities	Pollution issue
Wind Speed	High in storms	Climate impact

Key Conclusions

- Temperature varies significantly across seasons
- AQI indicates urban pollution issues
- Humidity correlates with rainfall
- Weather patterns differ by geography

8. Future Scope

- AI-based weather prediction
- Real-time dashboards
- IoT integration

9. References

- Power BI Documentation
- Weather datasets (Kaggle/OpenWeather)
- Data Analytics textbooks

10. Bibliography

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- Python for Data Analysis – Wes McKinney
 - Power BI Official Docs
 - Weather Data Research Papers

Linkedin Post Link - https://www.linkedin.com/posts/hemanth-gummadidala-617037333_powerbi-dataanalytics-datavisualization-ugcPost-7452782072596701184-aWVe?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAFPVFbMBjhJQNC8L39Z2JXIbU_rDRZqBWY_E